

■ AmritChidiya (■■■■■■■■■■) — Comprehensive Technical Architecture Handbook

Tagline: "Apni Sone Ki Chidiya Ko Phir Se Udaan Do" ■■

Mission: Empowering every Indian citizen with instant, AI-guided access to official government schemes, scholarships, and cyber safety tools in their native language through voice and text.

Target Audience: From fresh students and first-time coders to senior software architects and academic evaluators.

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1. Executive Summary & The Real-World Problem (Explained for Beginners)

Every year, the Government of India and state governments budget tens of thousands of crores of rupees for student scholarships, farmer income support, women empowerment programs, and healthcare subsidies. Yet, a vast percentage of eligible citizens never receive a single rupee. Why?

- 1. Linguistic Exclusion:** Official government portals are predominantly in complex bureaucratic English or high formal Hindi. A rural villager or regional student speaking Marathi, Tamil, or everyday Hinglish cannot understand the application requirements.
- 2. Opaque Multi-Factor Eligibility:** A student's eligibility depends on multiple intersecting criteria: Annual Family Income, State Domicile, Caste Category (General, OBC, SC, ST), Academic Year (10th pass, 12th pass, B.Tech), and Gender. Finding which exact scheme matches their profile requires days of manual research.
- 3. The Last-Mile Digital Divide:** Millions of citizens in rural villages do not own laptops or printers. They must visit physical Common Service Centers (Jan Seva Kendra / Cyber Cafes) where operators often lack complete information or charge unnecessary fees.
- 4. Rampant Cyber Fraud & Phishing:** Fraudulent WhatsApp forwards claiming *"Free Laptop Scheme 2026! Pay ₹299 to register"* cheat vulnerable families out of their hard-earned money.

AmritChidiya (■■■■■■■■■■) solves all four bottlenecks:

- Acts as a caring, multilingual companion that listens and speaks in **5 Indian languages** (Hindi, Hinglish, English, Marathi, Tamil).
- Uses an anti-hallucination AI engine and dynamic calculator to evaluate eligibility and estimate annual cash benefits.
- Protects citizens with a built-in **Cyber Shield** verifying official **.gov.in** and **.nic.in** domains.
- Outputs a physical, printable **CSC Scheme Dossier** with an NPCI bank checklist and an official operator stamp box for village cyber cafes.

2. The Complete Citizen Journey: Step-by-Step Walkthrough

Here is the exact step-by-step user journey from the moment a citizen opens AmritChidiya:

```
sequenceDiagram
    autonumber
    actor Citizen as Citizen
    participant FE as Member 1: Frontend (Next.js 16)
    participant BE as Member 2: Backend (FastAPI)
    participant Agent as Member 3: AI Agent & Safety (LangGraph)
    participant DB as Member 4: Database & Auth (Supabase/Bcrypt)

    Citizen->>FE: 1. Selects Language (Hindi, Hinglish, English, Marathi, Tamil)
    FE->>Citizen: Displays localized greeting, voice model & interface
    Citizen->>FE: 2. Speaks via Talk Mode or types inquiry
    FE->>BE: POST /transcribe (Audio Blob)
    BE->>BE: Groq Whisper Large v3 (with Indian phonetic bias)
    BE-->>FE: Returns clean Hindi/Hinglish text
    Citizen->>FE: 3. Opens Yojana Calculator & sets filters (State, Income, Profession)
    FE->>FE: Dynamically computes matched schemes & cash benefits
    Citizen->>FE: 4. Clicks "Get Step-by-Step Guide →"
    FE->>BE: POST /chat (Messages + Profile Context + JWT)
    BE->>BE: Validates JWT token & checks SlowAPI rate limits
    BE->>Agent: Invokes LangGraph State Machine
    Agent->>Agent: Runs Profile Intake & Scheme Match Engine
    Agent-->>BE: Returns response text & verified government portal URLs
    BE->>FE: Delivers JSON response with hyperlinked cards
    FE->>Citizen: 5. Displays matched schemes (e.g. UP Scholarship, PM-Kisan)
    Citizen->>FE: 6. Clicks "CSC Dossier"
    FE->>FE: Generates printable receipt (AC-CSC-XXXXXX) with NPCI bank checklist
    Citizen->>FE: 7. Pastes suspicious WhatsApp link into Cyber Shield
    FE->>Agent: Validates domain against .gov.in registry
    FE->>Citizen: Displays Verified Official Domain / Fraud Alert Badge
    FE->>DB: 8. Synchronizes encrypted chat history to Supabase Cloud
```

3. System Architecture Demystified (Explained with Analogies)

```
graph TD
    Client["Next.js 16 Frontend<br><i>The Citizen Counter</i>"]
    API["LangAPI Backend Server<br><i>The High-Speed Post Office</i>"]
    LLM["LangGraph + Groq LLM<br><i>The Scheme Expert Brain</i>"]
    STT["Groq Whisper API<br><i>The Listening Ears</i>"]
    TTS["Microsoft Edge-TTS<br><i>The Speaking Voice</i>"]
    DB["Supabase Cloud + SQLite Failover<br><i>The Bank Vault & Offline Backup</i>"]

    Client -->|HTTP POST /chat| API
    Client -->|HTTP POST /transcribe| API
    Client -->|HTTP POST /tts| API
    Client -->|HTTP /signup & /login| API

    API --> LLM
    API --> STT
    API --> TTS
    API --> DB
```

- Frontend (Next.js 16 + React 19) = "The Citizen Counter":

The interactive touch screen where citizens type, click dropdowns, view scheme cards, and trigger prints.

- **Backend API (FastAPI) = "The High-Speed Post Office":**

Receives all incoming requests, routes voice recordings, verifies digital passes (JWT), enforces rate limits, and returns answers in milliseconds.

- **AI Core (LangGraph + Groq LLM) = "The Scheme Expert Brain":**

Follows a strict state machine to prevent making up fake schemes and determines exact eligibility.

- **Speech Engine (Grog Whisper + Microsoft Edge-TTS) = "The Ears and Voice":**

Listens with phonetic understanding of Indian regional dialects and speaks back in warm, natural neural voices.

- **Database & Security (Supabase + SQLite + Bcrypt) = "The Bank Vault & Backup Safe":**

Encrypts passwords with 12 rounds of cryptographic salt and keeps an instant offline replica running 24/7.

4. 4 Engineering Pillars & Individual Member USPs

Each team member is the sole technical owner of one critical department:

Team Member	Engineering Role & Focus	Standout Member USP (Unique Selling Proposition)
Member 1	Frontend UI/UX & Citizen Access Lead	Phyigital Last-Mile Bridge & Citizen CSC Dossier Generator Bridges the digital-to-physical divide by generating structured, printable CSC Scheme Dossiers (AC-CSC-XXXXXX) with NPCI/DBT bank checklists & VLE operator sign-off boxes for village Jan Seva Kendras, coupled with a responsive Next.js 16 UI and hands-free Talk Mode.
Member 2	Backend API & Speech Processing Lead	Low-Latency Vernacular Audio Pipeline with Indian Accent DSP Engineered an end-to-end voice pipeline using Groq Whisper Large v3 with phonetic domain bias for Indian accents, ambient silence/hallucination suppression, and high-fidelity Microsoft Edge-TTS neural streaming in 5 Indian languages.
Member 3	AI Agent, Scheme Logic & Cyber Safety Lead	Anti-Hallucination Welfare Match Engine & Domain Cyber Shield Built the LangGraph dual-phase state machine that prevents welfare program hallucinations through strict intake rules, paired with the Official Domain Shield & Scam Checker that cryptographically verifies <code>.gov.in/.nic.in</code> domains to protect citizens from WhatsApp phishing.
Member 4	Database, Auth Security & DevOps Lead	Zero-Trust Hybrid Cloud Persistence with Bcrypt Salted Auth Architected an enterprise-grade dual-database architecture with Supabase Cloud PostgreSQL and instant local SQLite failover, hardened by salted Bcrypt (12 rounds) , signed 24-hour JWT tokens , SlowAPI rate limiting, and production CORS deployment.

5. Step-by-Step Technical Deep Dive for Every Role

■■■ Role 1: Frontend UI/UX & Citizen Access Lead (Member 1)

- **Concept in Plain Words:** Builds everything the citizen touches and sees, ensuring it runs smoothly on cheap smartphones and outputs physical paper receipts for rural citizens.
- **Step-by-Step Execution:**
 1. *State Management* (**page.tsx**): Coordinates conversational turns, language switches, speech audio states, and guest message limits.
 2. *Instant Yojana Calculator* (**EligibilityCalculatorModal.tsx**): Evaluates State, Category, Income, and Profession client-side in real-time, showing estimated annual benefit amounts (e.g. ₹30,000–₹55,000/yr). Clicking “*Get Step-by-Step Guide* →” passes user context directly to the chat companion.
 3. *Phygital CSC Dossier Engine* (**CscDossierModal.tsx**): Generates an official receipt (**AC-CSC-XXXXXX**) with applicant details, verified schemes, an NPCI/DBT bank checklist, and a physical VLE Operator stamp box with custom **@media print** CSS for standard A4 paper.

